Mechatronic Actuators Trainer

User Manual – Setup and Configuration

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For more information on the solutions Quanser offers,
please visit the web site at: http://www.quanser.com

Quanser Consulting Inc.
119 Spy Court
Markham, Ontario
L3R 5H6, Canada

info@quanser.com
Phone : 19059403575
Fax : 19059403576
printed in Markham, Ontario.

Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including een tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.

Japan VCCI Notice This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI). If this equipment is used in a domestic environment, radio interference may occur, in which case the user may be required to take corrective actions.

この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。 VCCI–A

Waste Electrical and Electronic Equipment (WEEE)
This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

电子信息产品污染控制管理办法 (中国 RoHS)

中国客户 Quanser Consulting Inc. 关于关于限制在电子电气设备中使用某些有害成分的指令 (RoHS)。

CE Compliance

This product meets the essential requirements of applicable European Directives as follows:
- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Caution

Warning - Hot Surface! Surfaces of BLDC and Stepper motors can exceed 50˚C/120˚F during prolonged operation. Do not touch motors until they have cooled down.

Table of Contents

A. Introduction

The Mechatronic Actuators Trainer is an integrated motor experiment designed for undergraduate mechatronics and related courses. It is designed to help teach fundamental control concepts and theories on an easy-to-use and intuitive platform.

The compact device provides an intuitive solution that can drive and sense four motor types: brushed DC, brushless DC (BLDC), stepper, and servo motors, while providing rotational position and speed feedback and per-channel current sensing. The system emphasizes hands-on learning: actuator fundamentals, hardware interfacing, motor selection, control, and simple transmission mechanisms.

This allows your students to learn the fundamental workings of these actuators and develop key skills in motor selection based on functionality and performance requirements. This solution is ideal for teaching undergraduate mechatronics, and is accompanied by comprehensive fundamentals labs, integrated challenges, as well as relevant industry open-ended projects.

The unit is intended for lab and classroom use to support fundamentals labs, guided integration challenges, and open-ended projects.

Figure 1. Mechatronic Actuators Trainer

B. Hardware Components

The main components of the Actuators Trainer are listed in Table 1 and 2. These components are marked in Figures 2 and 3, which show the front, back and side views of the Trainer.

The datasheet for each of the components is located in the same folder as this user manual.

i. Parts of the Device

Figure 2. Front of Mechatronic Actuators Trainer

ID	Component	ID	Component
1	Block Section	5	Brushless DC Motor Connector
2	Block Status LED	6	5 pin Encoder Connector
3	Brushed DC Motor Connector	7	Stepper Motor Connector
4	Servomotor Connector		

Table 1. Components on the front of the device

Figure 3. Side of Mechatronic Actuators Trainer

ID	Component	ID	Component
8	USB Status LED	11	Power Connector
9	USB C Data Port	12	Power Switch
10	Power Status LED		

Table 2. Components on the side of the device

ii. External Provided Parts

The Mechatronic Actuators Trainer includes four actuators, one for each type of motor that can be driven with the device. Each motor connects directly to one of the device's motor channels via dedicated connectors and cables. The datasheets from each motor are located in the same folder as this user manual. Extra connectors are provided to allow using different motors with the device. Motor hub attachments are also provided for 3 of the D-Shaft motors and a servo horn set for the servo standard shaft. The following picture shows the provided power supply, usb cables and the extra connectors and pieces provided.

iii. Included Actuators

Four different actuators are included with the device.

Overview of Actuators			
Actuator Type	Model	Description	Shaft Type
Brushed DC Motor	GMP24-370CA	Compact, brushed DC motor with integrated gearbox. Provides speed control.	5 mm D-Shaft
Brushless DC Motor	PBLR32HES-121	Three-phase brushless DC motor with integrated Hall sensors for commutation and precise speed sensing.	5 mm D-Shaft
Servo Motor	DMOND DCS16LP	Digital servo with high torque and integrated feedback electronics for angular position control.	Std. servo shaft (servo spline)
Stepper Motor	NEMA 17 (PHB42S series)	Hybrid stepper motor with 1.8° step angle for precise incremental motion.	5 mm D-Shaft

Table 3. Overview of Included Actuators

a. Brushed DC Motor - GMP24-370CA

The GMP24-370CA is a brushed DC motor with an integrated precision planetary gearbox. It features bidirectional control through PWM and includes an integrated 24:1 gearbox that increases torque output while maintaining smooth and predictable speed characteristics.

GMP24-370CA	
Parameter	Specification
Rated Torque	5 KG.CM
Gear Reduction	24 : 1
Nominal Voltage	12 V

Table 4. Brushed DC Motor Specs

b. Brushless DC Motor – PBLR32HES-121

The PBLR32HES-121 Brushless DC Motor is a three-phase motor equipped with integrated Hall sensors for commutation and speed feedback. The motor features a compact 32 mm frame size, low cogging torque, and consistent speed characteristics.

PBLR32HES-121	
Parameter	Specification
Rated Torque	0.083 N·m
Hall Sensors	3 sensors (120° spacing)
Nominal Voltage	12 V

Table 5. Brushless DC Motor Specs

c. Servo Motor – DCS16LP

The DMOND DCS16LP Digital Servo Motor is a high-torque servo equipped with a metal geartrain and dual ball bearings. It supports a wide operating voltage range from 4.8 V to 8.4 V, providing stable performance under load. The servo features a PWM control interface with a 500–2500 μs pulse width range and a total rotation angle of approximately 180°.

DCS16LP	
Parameter	Specification
Stall Torque	12 – 16.5 kg·cm
No-Load Speed	0.10 – 0.07 s / 60°
Operating Voltage	4.8V – 8.4V DC

Table 6. Servo Motor Specs

d. Stepper Motor – NEMA 17 (PHB42S Series)

The NEMA 17 PHB42S Series Stepper Motor is a two-phase hybrid stepper motor with a 1.8° step angle. It offers high torque output and consistent step accuracy suitable for general motion control applications.

DCS16LP	
Parameter	Specification
Step Angle	1.8° ± 5%
Holding Torque	0.52 N·m
Rated Current	1.7 A/phase

Table 7. Stepper Motor Specs

iv. Encoder Standards

Encoder outputs are assigned to match the standard across Quanser devices, which could lead to a mismatch with the datasheet. The standard in Quanser devices is B leads A for a CW direction. A positive motor input leads to a counterclockwise rotation of the shaft which increases the encoder counts. This matches all previously released Quanser legacy devices.

v. Capabilities

The Quanser Mechatronic Sensors Trainer provides the following capabilities:

- PWM commands to control four brushed DC motors
- PWM commands to control four brushless DC motors
- PWM commands to control four servo motors
- PWM commands to control four stepper motors
- Built-in device protection with current monitoring and limiting
- Four external encoders (with 2X and 4X quadrature)

C. Handling and Setup

i. Powering Up the Device

To set up the Mechatronic Actuators Trainer, follow these steps:

1. Make sure the Mechatronic Actuators Trainer is out of its case and placed on a stable surface.
2. Power the device up using the provided Power Supply or the cables provided to use an external battery. If powering up with something other than the provided power supply, please refer to the Power Supply Specifications section of this user manual to verify the allowed voltages and currents.
3. Connect the device to a computer (PC with windows or Linux or a Raspberry Pi) using the USB cable provided.
4. Power on the device using the Power Switch next to the Power connector.
5. The power LED, USB LED and blocks LEDs should all be green. If they are not green, refer to the LED Status section of this user manual.
6. The Mechatronic Actuators Trainer is now ready to be used.

ii. Power Supply Specifications

The Mechatronic Actuators Trainer operates from a central power supply shared across all actuator blocks (0–3). Individual blocks do not have separate power ratings.

The provided power supply is capable of 12V 8.3A.

If you use anything else to power the device, the power supply thresholds are as follows:

Parameter	Absolute Limit	Default Setting
Minimum Voltage	8 V	10V
Maximum Voltage	24 V	16V
Maximum Current	10 A	8A

Note: Default thresholds can be adjusted in the file that will be run. Absolute limits represent the extreme values that can be set safely.

iii. Errors

The Mechatronic Actuators Board has various safety checks to verify that the device is not pushed past its electrical limits. The errors that the device can encounter are in the following table. Refer to the LED Status section for more information about the LED states for each of the status LEDs.

Error	Threshold (absolute)	Threshold (default)	LED State
Power Supply Voltage Too Low (V)	8 V	10V	Power - Red
Power Supply Voltage Too High (V)	24 V	16V	Power - Red
Power Supply Current Too High (A)	10 A	8A	Power - Red
DC Motor Current Too High (A)	3 A	1.1 A	Block - Red
Servo Supply Current Too High (A)	3 A	1.3 A	Block - Red
BLDC Current Too High (per channel) (A)	6 A	2 A	Block - Red
Stepper Current Too High (per winding) (A)	3 A	1.7 A	Block - Red

iv. Driver Connector Voltages

The connectors in each of the blocks have different voltages. These voltages are shown in the table below. No matter the voltage input to the device, the encoder and servo outputs will always output 5V to the devices. The dc motor, bldc motor and stepper outputs are equal to the power supply input voltage.

Note that sending a 1 in the code will give a different voltage rating depending on the voltage input of the power supply.

Connectors	Voltage
Encoder	5 V (fixed)
Servomotor	5 V (fixed)
DC Motor	Equal to power supply input voltage
BLDC Motor (per channel)	Equal to power supply input voltage
Stepper motor (per winding)	Equal to power supply input voltage

v. LED Status

The Mechatronic Actuators Trainer has 6 non-configurable LEDs. One per each block of motors (4 total), a Power LED and a USB LED.

These LEDs help troubleshoot the status of the system. The color codes are as follows:

Block LEDs:

- Green: Everything is good/working
- Blue: Watchdog timer expired
- Red: overcurrent detected on one of the motor outputs. NEED MOTOR REENABLE OR RESTARTING CODE.
- Off: DFU (flashing) mode. ONCE DEVICE IS FLASHED, IT NEEDS TO BE RESET.

Power LED:

- Green: Device is powered on and everything is good/working
- Red: Over/under-voltage, or over-current detected on main power supply input. See Power Supply Specifications section. NEEDS DEVICE RESET.
- Orange: Over-temperature inside the device case.
- Off: Device is powered off.

USB LED:

- Green: USB cable detected.
- Off: No USB cable detected.

vi. Dimensions

Dimension	Size
Length	18 cm
Width	3.6 cm
Height	13 cm
Weight	290 grams

D. Environmental

The Mechatronic Actuators Trainer is designed to function under the following environmental conditions:

- Standard rating
- Indoor use only
- Atmospheric conditions
    - Temperature 5∘C to 40∘C
    - Altitude up to 2000m
    - Maximum relative humidity of 80% up to 31∘C decreasing linearly to 50% relative humidity at 40∘C
- Pollution Degree 2
- Mains supply voltage fluctuations up to 10% of the nominal voltage
- Maximum transient overvoltage 2500V
- Marked degree of protection to IEC 60529: Ordinary Equipment (IPX0)

E. Electrical Considerations

⚠	Caution	If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.

⚠	Caution	Surface of motors may become hot during prolonged operation.

⚠	Caution	The Mechatronic Actuators Trainer is not waterproof.

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